

Customer Churn Analysis

Understanding who leaves, why they leave, and how much it costs — using data to drive smarter retention decisions.



What We Set Out to Solve

Demographics & Tenure

Age, gender, and how long customers stay

Engagement & Support

Usage frequency, last interaction, and support calls

Payment & Contracts

Payment delays, subscription type, and contract length

Business Impact

Revenue loss and high-risk segment identification

Dataset at a Glance

440K

Customer Records

After removing one empty row; no duplicates found

12

Features

Demographics, tenure, engagement, support, payment, subscription, contract, spend, churn

56.71%

Churn Rate

249,999 churned vs. 190,833 retained customers

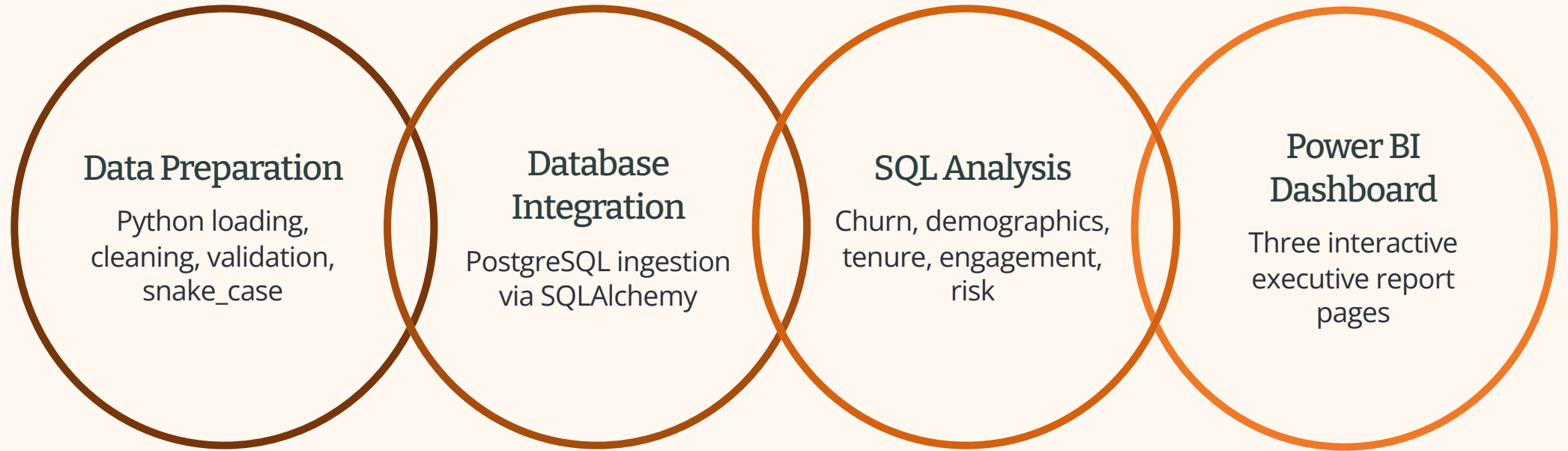
43.29%

Retention Rate

Overall retention across the full customer base

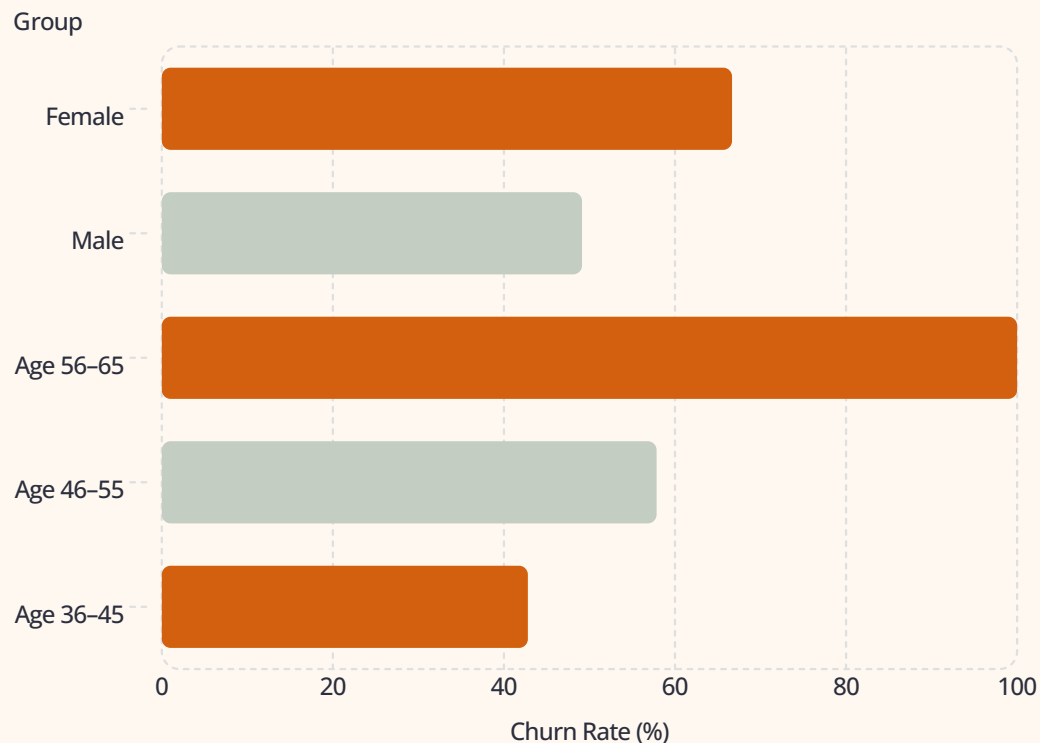
📌 The high churn rate makes customer retention a critical business priority.

Analysis Pipeline: Python → SQL → Power BI



The cleaned dataset was loaded into a `customer_churn` table in PostgreSQL for structured SQL analysis, then visualized in an interactive Power BI dashboard.

Who Churns More? Demographics & Tenure



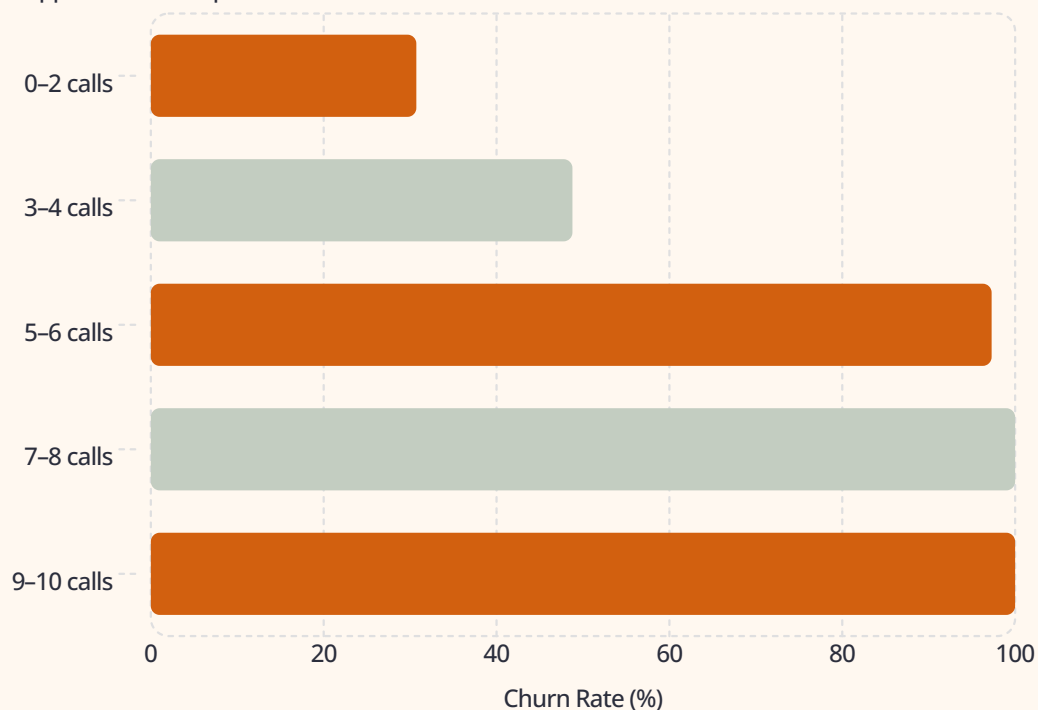
Key Findings

- **Female customers** churned at **66.67%** vs. 49.13% for males
- **Age 56-65** had a 100% observed churn rate — highest of all age groups
- **13-24 month tenure** group led tenure segments at 63.27% churn
- Average tenure: Churned 30.47 mo vs. Retained 32.28 mo

📌 These are associations in the dataset — not evidence of direct causation.

Behavioral Signals: Engagement & Support

Support Calls Group



Support Calls

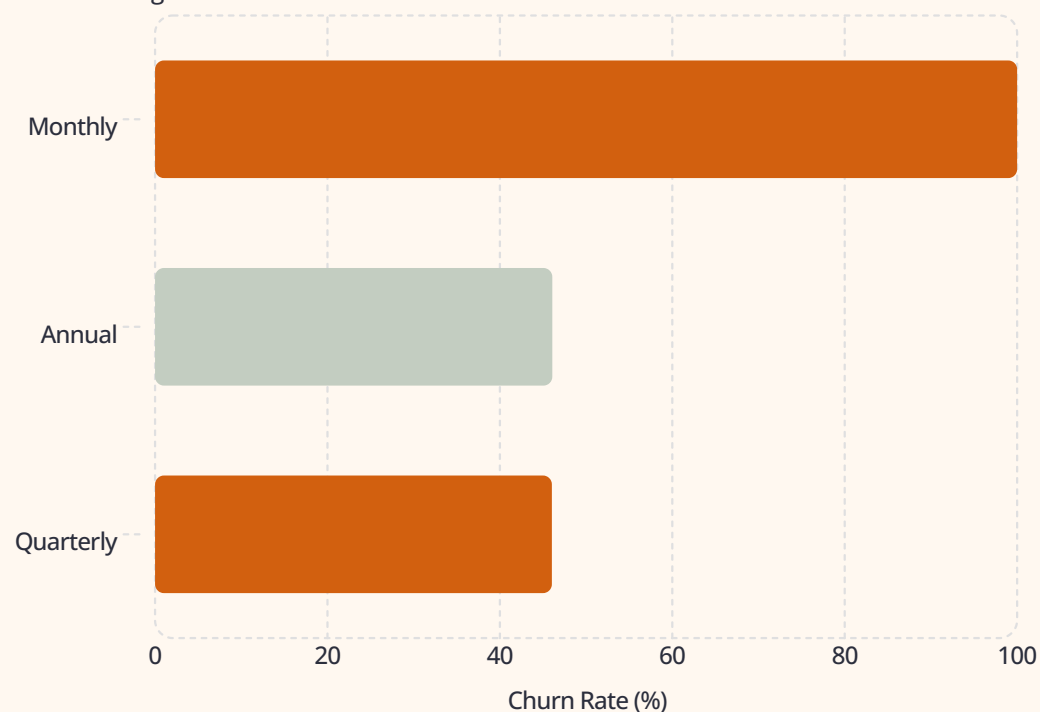
Churned customers averaged **5.14 calls** vs. 1.59 for retained. Customers with 7-10 calls had a 100% churn rate.

Engagement

- Lowest usage (1-5 events): **61.50%** churn rate
- 16-30 days since last interaction: **~66%** churn vs. ~49% for 1-15 days

Payment Delays & Contract Type

Contract Length



Payment Behavior

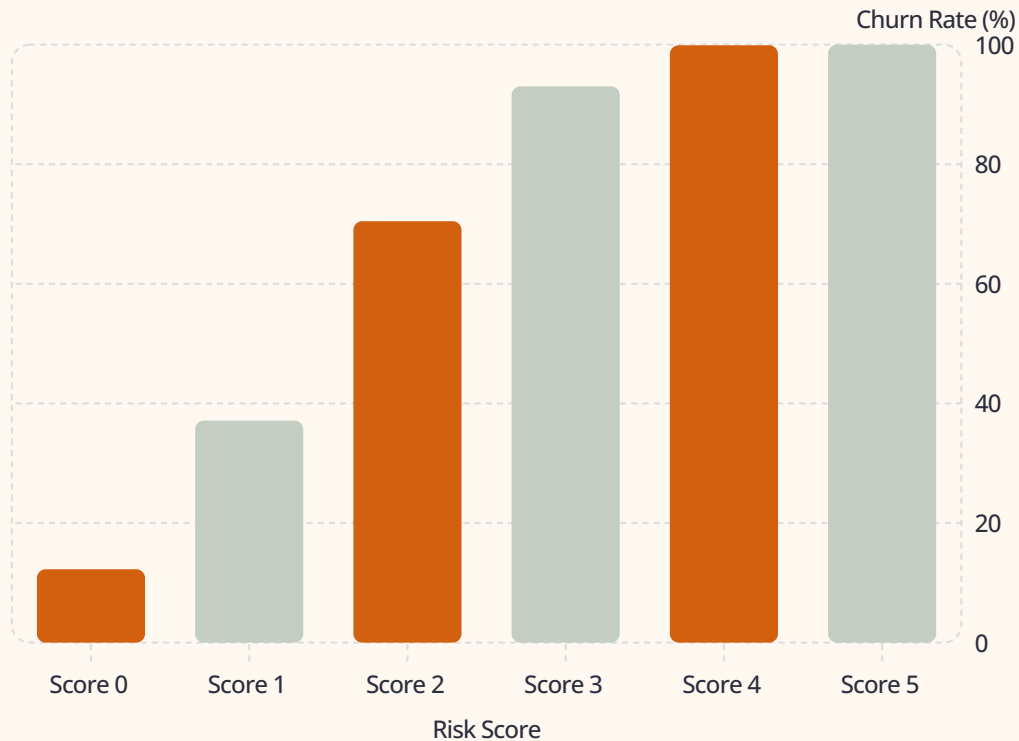
Churned customers averaged **15.22 days** payment delay vs. 10.02 days retained. Delays of 21–30 days showed 100% churn.

Subscription Type

Basic led at 58.18%, but differences were small. **Contract length** was the far stronger signal — monthly contracts at 100% churn.

High-Risk Customer Segments

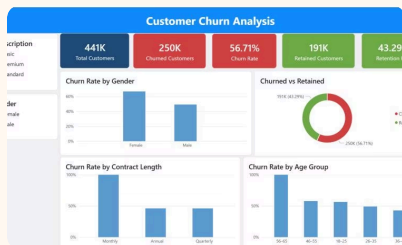
Risk Score (0–5) combines contract length, support calls, payment delay, and engagement. Churn rises sharply with each added risk indicator.



Top-Risk Combinations

- **Monthly contracts:** Multiple combinations at 100% churn regardless of other factors
- **Quarterly/Annual + High Support + High Payment Delay + Low Usage:** 99.50–99.56% churn
- **109K high-risk customers** identified with a 95.01% combined churn rate

Three-Page Interactive Dashboard



Page 1 — Executive Overview

High-level KPIs: total customers, churn rate, retention rate, and breakdowns by gender, age, and contract length.



Page 2 — Behavior & Churn Drivers

Churn rates by usage frequency, last interaction, support calls, and payment delay.



Page 3 — Risk & Business Impact

Risk scores, high-risk segments, churned customer spend (\$135.32M lost), and financial impact by risk level.

Key Recommendations

→ Prioritize by Risk Score

Use the 0–5 Risk Score to focus retention on highest-risk customers first

→ Address Support & Payment Issues

Investigate customers with 5+ support calls; offer proactive payment-support programs

→ Re-engage Low-Usage Customers

Targeted offers and product education to increase usage and reduce inactivity

→ Convert Monthly Contracts

Monthly contracts show 100% churn — prioritize upgrades to annual or quarterly plans

📄 Overall churn: **56.71%**. Risk Score 0 → 12.27% churn; Score 5 → 100%. Use the Power BI dashboard for ongoing monitoring.

